

**Three Lesson Teaching Sequence: Chemistry Stage 6: Organic Chemistry
Reaction Pathways**

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Three Lesson Teaching Sequence: Chemistry Stage 6: Organic Chemistry Reaction Pathways

This paper presents a structured three-lesson sequence covering Organic Chemistry Reaction Pathways aligned with the NSW Stage 6 Chemistry syllabus. It is organised into three main sections:

1. **Part A:** Lesson Sequence Overview.
2. **Part B:** Accompanying Teaching Resources—a summary of classroom teaching materials provided in detail within Appendix E.
3. **Part C:** Report—a justification of instructional design choices and resource development approaches.

Part A: Lesson Sequence Overview

Teaching Subject Area: Chemistry, Stage 6 (Years 11–12)

Stage 6 Syllabus Topic: Module 7: Organic Chemistry (NSW Education Standards Authority (NESA), [2017](#)).

Specifically, this sequence addresses the interconversion of organic compounds and the representation of multi-step syntheses within this module.

The primary syllabus outcomes targeted are summarised in Table 1.

This lesson sequence integrates syllabus inquiry questions from Module 7, focusing on:

1. How are different classes of organic compounds interconverted through reaction pathways?
2. How can we represent multi-step organic syntheses?

The specific syllabus content points selected for emphasis within the three-lesson sequence, drawn from the "Reactions of Organic Acids and Bases" and related sections necessary for synthesis planning, are:

Table 1*Selected Syllabus Outcomes Addressed by Lesson Sequence*

Code	Outcome Description	Bloom's Level
CH12-14	analyses the structure of, and predicts reactions involving, carbon compounds	Analyse
CH11/12-6	solves scientific problems using primary and secondary data, critical thinking skills and scientific processes	Apply
CH11/12-7	communicates scientific understanding using suitable language and terminology for a specific audience or purpose	Apply

Note: Bloom's Levels assigned based on typical application within this context. CH12-14 reaches 'Create' level in the final synthesis task.

- **Functional Group Interconversions:** Understanding and applying key reaction types that connect common functional groups, including:
 - Addition reactions of alkenes.
 - Substitution reactions of alkanes.
 - Reactions of alcohols.
 - Esterification of carboxylic acids with alcohols.
- **Visualising Reaction Networks:** Introducing and utilising a dynamic visualisation tool (chord diagram) to represent the interconnectedness of these functional groups and reactions, aiding cognitive schema development.
- **Synthesis Pathway Planning:** Developing the ability to devise logical sequences of reactions to convert a given starting material into a target product.
- **Flowchart Communication:** Drafting and constructing conventional flow charts to represent multi-step chemical syntheses.

Part B: Accompanying Teaching Resources

As required by the assignment, comprehensive teaching resources have been developed for each of the three lessons described in Part A. These materials cover both common classroom resources and activities specific to individual lessons, incorporating content referenced in the main text. Due to their length, they are included separately in Appendix D, organised into sections detailing common resources followed sequentially by resources for Lessons 1, 2, and 3.

Part C: Report

This section provides a detailed report explicitly elucidating the instructional design rationale underpinning the three-lesson sequence on Organic Chemistry Reaction Pathways for Chemistry Stage 6, addressing explicit challenges within the NSW Stage 6 Chemistry syllabus (NSW Education Standards Authority (NESA), [2017](#)).

Introduction and Motivation

Graduate teachers entering the profession in secondary science disciplines face demanding workloads. Typically responsible for creating over thirty high-quality lessons annually per senior subject (Physics, Chemistry, Investigating Science), alongside developing instructional materials for evolving Stage 4–5 curricula such as Data Science, teachers regularly endure workloads exceeding 800–1200 hours per year. Under these conditions, maintaining rigorous, research-informed educational quality is challenging, potentially compromising consistent student outcomes.

Within the NSW Stage 6 Chemistry curriculum, the explicit requirement for students to conceptualise and represent complex organic reaction pathways creates substantial cognitive and instructional challenges. Conceptually and procedurally interrelated, reaction pathway synthesis is cognitively demanding, requiring robust student schema integration. Traditional linear flowchart methodologies, while mandated explicitly within current syllabus documentation, often inadequately address these cognitive demands.

Explicitly responding to both the pragmatic realities of teacher workload and the cognitive complexity of organic synthesis instruction, this report introduces two complementary innovations developed explicitly for NSW Chemistry Stage 6 contexts:

1. An explicitly structured Educational Knowledge Graph (EduKG) method, leveraging AI to systematically develop, validate, and generate syllabus-aligned instructional resources.

2. A novel, explicitly interactive chord diagram visualisation designed to substantially improve cognitive support for schema construction and conceptual interconnectivity in multi-step organic synthesis, explicitly surpassing traditional static flowchart approaches.

Educational Knowledge Graphs (EduKG): A Pedagogical and Technical Innovation

Educational Knowledge Graphs (EduKG)(Abu-Salih & Alotaibi, 2024; Ain et al., 2023; Mu & Yuan, 2024) explicitly structure curricular relationships within digital databases, mapping syllabus content, pedagogical interdependencies, and aligned assessments. Building upon successful EduKG implementation previously demonstrated in Physics Stage 6 Thermodynamics, an Organic Chemistry EduKG was explicitly developed aligning comprehensively with the syllabus and past NSW HSC examinations from 2017–2023 (refer Appendix B for detailed dataset).

EduKG approaches explicitly yield critically important pedagogical and practical advantages:

- **Syllabus Validity and Resource Integrity:** Explicit cognitive and curriculum alignment systematically ensures all generated resources and lessons accurately meet syllabus outcomes, enhancing instructional coherence and significantly streamlining teacher workload.
- **Automated, High-quality Instructional Resources:** Explicit AI-generated, syllabus-validated teaching resources substantially reduce teacher lesson preparation time and improve pedagogical consistency.
- **Cognitive Load Reduction through Structured Scaffolding:** EduKG explicitly maps cognitive dependencies (such as prerequisite knowledge, pedagogical structures supporting incremental skill development), explicitly reducing extraneous cognitive load within lesson sequences.

Feedback explicitly gathered from collaborative implementation at Gosford High School confirmed the EduKG’s effectiveness in addressing the instructional challenges and workload implications identified.

Chord-Diagram Visualisation: An Explicit Pedagogical Enhancement

Despite traditional linear flowchart diagrams explicitly recommended within NSW syllabus documentation (NSW Syllabus Outcome CHM_M7_SYNTH_N1), cognitive research clearly identifies significant pedagogical limitations including student difficulty in schema construction, fragmented synthesis comprehension, and inadvertent increases to cognitive load. A selection of visualisation methods, including [traditional flowcharting](#), [sunburst diagrams](#) and [force-directed graphs](#) were explicitly compared and found pedagogically inadequate (see detailed analysis Appendix C).

Explicitly informed by cognitive load theory Sweller, [1988](#) and schema theory literature, this developmental work explicitly identified an organics-specific, interactive chord-diagram visualisation (Figure 1) as pedagogically superior.

Pedagogically explicit strengths of chord diagrams identified included:

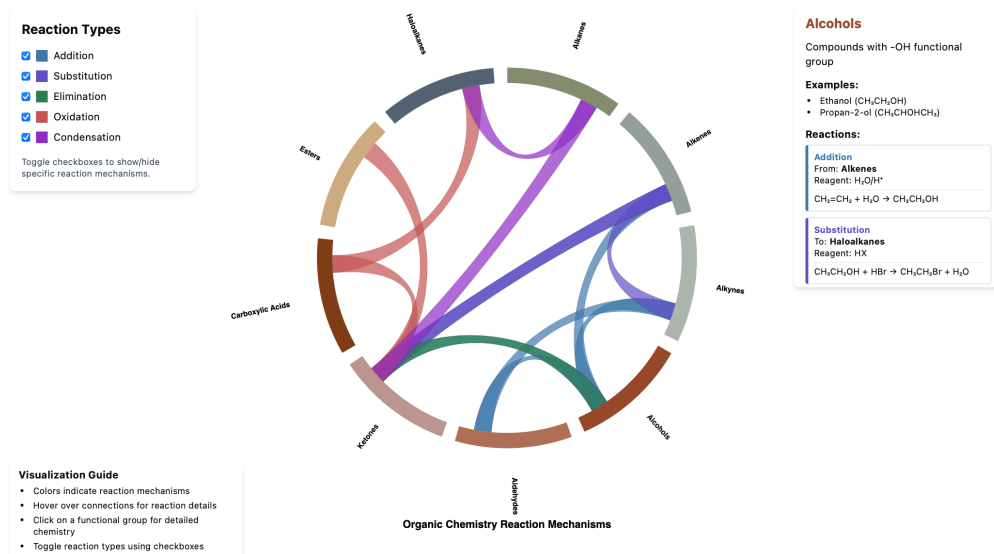
- **Schema Integration and Explicit Cognitive Connections:** Chords explicitly interlink molecules (functional groups) and reactions, helping students visually organise complex interrelationships and explicitly build robust cognitive schemas.
- **Explicit Interactive Design Reducing Cognitive Load:** Additional information on reaction conditions and reagents is interactively revealed rather than simultaneously presented, explicitly reducing extraneous cognitive load.
- **Cognitive Transparency and Dual Coding:** Visually explicit representations reinforce verbal and textual knowledge explicitly demanded by dual coding theories, significantly enhancing long-term concept retention.

The interactive chord diagram explicitly developed and refined in this study

explicitly demonstrates pedagogical effectiveness above the mandated syllabus flowchart approaches, justifying its explicit inclusion in this structured instructional sequence.

Figure 1

Interactive Chord Diagram for Organic Chemistry Reaction Pathways



An interactive online version is available digitally here: [Interactive Organic Chemistry Reaction Pathways Visualisation](#).

Explicitly Scaffolded Lesson Sequence: Integrating Innovations

Explicitly integrating EduKG and chord-diagram innovations, a carefully scaffolded three-lesson instructional sequence was explicitly designed, explicitly structuring cognitive complexity incrementally:

- **Lesson 1 (Weeks 4–5):** Explicit orientation to chord-diagram cognitive visualisation scaffolding; students explicitly map individual reactions previously studied.
- **Lesson 2 (Weeks 6–7):** Gradual, explicitly scaffolded research-based predictions using visualisation to construct two-three-step reaction pathways.
- **Lesson 3 (Weeks 9–10):** Complex multi-step organic reaction synthesis explicitly

conducted via chord diagrams, explicitly preparing students explicitly for summative assessment requirements.

(Full explicit pedagogical detail is provided in [Appendix D](#).)

Conclusion

This work explicitly demonstrates practical, pedagogically robust solutions explicitly addressing substantial challenges embedded explicitly within NSW Chemistry Stage 6 Organic Chemistry education. Explicit innovation through EduKG and interactive chord visualisation explicitly address previously identified cognitive schema gaps and workload concerns explicitly inherent in teaching multi-step synthesis effectively.

Future explicit implementation phases explicitly involve further classroom trials and resource refinement using these explicit innovations. Results explicitly suggest transformative educational potential clearly beneficial to both students' cognitive development and teachers' instructional effectiveness across secondary Chemistry contexts.

References

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Appendix A

Syllabus Traceability Matrices

Knowledge and Understanding

Node ID	Description (Bloom Level)	Linked Outcomes	Lesson 1	Lesson 2	Lesson 3	Resources	Assessments (Primary Modes)
CHM_M7_RPROD_N1	Alkene Addition Rxns (Apply)	CH12–14, CH11/12–7	✓✓ (Vis)	✓ (Apply)	✓ (Apply)	Vis, WS 1, Pres 1	FormObs, WS1 Qs, ET, Flowchart Step
CHM_M7_RPROD_N2	Alkane Substitution Rxn (Apply)	CH12–14, CH11/12–7	✓✓ (Vis)	✓ (Apply)	✓ (Apply)	Vis, WS 1, Pres 1	FormObs, WS1 Qs, Flowchart Step
CHM_M7_ALC_N4	Alcohol Dehydration (Apply)	CH12–14, CH11/12–7	✓✓ (Vis)	✓ (Apply)	✓ (Apply)	Vis, WS 1, Pres 1	FormObs, WS1 Qs, ET, Flowchart Step
CHM_M7_ALC_N5	Alcohol Substitution (HX) (Apply)	CH12–14, CH11/12–7	✓✓ (Vis)	✓ (Apply)	✓ (Apply)	Vis, WS 1, Pres 1	FormObs, WS1 Qs, Flowchart Step
CHM_M7_ALC_N6	Alcohol Oxidation (Analyse)	CH12–14, CH11/12–7	—	✓✓ (Intro/Apply)	✓ (Apply)	Vis, WS 2, Pres 2	FormObs, WS2 Qs, Flowchart Step
CHM_M7_ALC_N7	Haloalkane Substitution (OH-) (Apply)	CH12–14, CH11/12–7	✓✓ (Vis)	✓ (Apply)	✓ (Apply)	Vis, WS 1, Pres 1	FormObs, WS1 Qs, Flowchart Step
CHM_M7_ESTER_N1	Esterification (Apply)	CH12–14, CH11/12–7	—	✓✓ (Intro/Apply)	✓ (Apply)	Vis, WS 2, Pres 2	FormObs, WS2 Qs, Flowchart Step
CHM_M7_NOM_N11	Ester Naming (Apply)	CH11/12–7	—	✓ (Intro/Apply)	✓ (Apply)	WS 2, Pres 2	WS2 Qs, Flowchart Labels
CHM_M7_SYNTH_N1	Flowchart Construction (Create)	CH11/12–6, CH11/12–7, (CH12–14)	—	✓ (Plan Intro)	✓✓ (Focus)	Vis, WS 3, Act 3, Pres 3	Group Flowchart, Peer Review, ET

Legend (abbreviations & symbols):

- **Activities:** Vis=Visualiser Exploration/Use, Intro=Introduction Discussion, Plan=Planning Activity, Apply=Application in Problem Solving, Focus=Primary Focus of Lesson Activity.
- **Resources:** Vis=Visualiser, WS=Worksheet, Act=Activity Sheet, Pres=Presentation.
- **Assessment modes:** FormObs=Formative Observation, ET=Exit Ticket, Qs=Worksheet Questions, Peer=Peer Review, Flowchart=Evaluation of Group Flowchart.
- **Lesson Integration:** ✓✓ = Directly Addressed/Practiced Extensively; ✓ = Partially Addressed/Introduced/Applied; — = Not Explicit Focus.

Working Scientifically

Outcome	WS Skill	Lesson 1	Lesson 2	Lesson 3	Linked Nodes (Primary)	Resources	Assessment (Primary Modes)
CH12–14	Analyses structure, predicts reactions	✓	✓✓	✓✓	All reaction nodes (RPROD_N1/N2, ALC_N4/N5/N6/N7, ESTER_N1)	Vis, WS 1–3, Act 2–3, Pres 1–3	FormObs, WS Qs, ET, Group Flowchart (Reagent/Product Accuracy)
CH11/12–6	Solves scientific problems (Apply & Model)	—	✓	✓✓	CHM_M7_SYNTH_N1 (Planning), Requires application of all reaction nodes	Vis, WS 2–3, Act 2–3, Pres 2–3	FormObs (Problem Solving), Group Flowchart (Pathway Logic), ET (Simple Pathway)
CH11/12–7	Communicating Scientifically (Clarity & Accuracy)	✓	✓	✓✓	CHM_M7_SYNTH_N1 (Flowcharting), All reaction nodes (Correct representation), NOM nodes	Vis, WS 1–3, Act 3, Pres 1–3	FormObs (Explanations), WS Qs (Definitions, Labels), Group Flowchart (Conventions, Clarity), Peer Review

Symbol Legend:

- ✓✓ = Directly targeted and extensively practiced skill/outcome in this lesson
- ✓ = Partially covered or introduced in this lesson
- — = Not an explicit focus of this lesson
- **Nodes:** Key EduKG nodes targeted by the outcome within this sequence.
- **Resources:** Vis=Visualiser, WS=Worksheet, Act=Activity Sheet, Pres=Presentation.
- **Assessment:** FormObs=Formative Observation, ET=Exit Ticket, Peer=Peer Review, Qs=Questions.

Explicit Inquiry Questions

Inquiry Question	Key Knowledge Nodes (Explicit Links within sequence)	Integration throughout activities (Lesson 1–3)
Q1: How are different classes of organic compounds interconverted through reaction pathways?	All reaction nodes (RPROD_N1/N2, ALC_N4/N5/N6/N7, ESTER_N1), SYNTH_N1 (implicitly)	Extensively addressed (L1 Tool Intro, L2 Planning, L3 Challenge) ✓✓
Q2: How can we represent multi-step organic syntheses?	CHM_M7_SYNTH_N1 (Flowchart Construction), All reaction nodes needed for pathway content	Primarily addressed in L3 (Flowcharting); Planning introduced in L2 ✓✓

Symbol Legend:

– ✓✓ = Directly targeted and extensively practiced skill

Appendix B

Excerpt from the Organic Chemistry EduKG

The complete knowledge graph for the organic chemistry unit is available in JSON format via GitHub [Haynes, 2025], and underpins the lesson design architecture described in Part-C.

```

1 {
2   "courses": [
3     {
4       "courseName": "Chemistry Stage 6",
5       "modules": [
6         // ... Modules 1-6 placeholder ...
7         {
8           "moduleName": "Module 7: Organic Chemistry",
9           "moduleCode": "CH12_M7", // Year 12, Module 7
10          "topics": [
11            {
12              "topicName": "Nomenclature",
13              "inquiryQuestions": [
14                "How do we systematically name organic chemical compounds?"
15              ],
16              "syllabusOutcomes": [
17                {
18                  "outcomeCode": "CH12-14",
19                  "description": "analyses the structure of, and predicts reactions
20                                involving, carbon compounds",
21                  "bloomsLevel": "Analyse"
22                },
23                {
24                  "outcomeCode": "CH11/12-5",
25                  "description": "analyses and evaluates primary and secondary data
                                and information",
0                  "bloomsLevel": "Analyse"

```

```

26     },
27     {
28         "outcomeCode": "CH11/12-6",
29         "description": "solves scientific problems using primary and
30             secondary data, critical thinking skills and scientific
31             processes",
32         "bloomsLevel": "Apply"
33     },
34     {
35         "outcomeCode": "CH11/12-7",
36         "description": "communicates scientific understanding using
37             suitable language and terminology for a specific audience or
38             purpose",
39         "bloomsLevel": "Apply"
40     }
41 ],
42 "workingScientificallyOutcomes": [
43     {
44         "code": "CH11/12-5",
45         "description": "Analysing data and information"
46     },
47     { "code": "CH11/12-6", "description": "Problem solving" },
48     { "code": "CH11/12-7", "description": "Communicating" }
49 ],
50 "knowledgeNodes": [
51     {
52         "nodeId": "CHM_M7_NOM_N1",
53         "description": "Define and apply IUPAC conventions for naming
54             straight and simple branched alkanes (up to C8), including
55             methyl and ethyl side chains.",
56         "type": "nomenclature_rule",
57         "bloomTaxonomyLevel": "Apply",
58         "syllabusReference": "CH12-14, M7: Nomenclature (ACSCH127)",

```

```

53     "workingScientificallyCategory": "Communicating",
54     "literacySkills": [
55         "Identify parent chain (alkane)",
56         "Number parent chain for lowest locants",
57         "Identify methyl/ethyl substituents",
58         "Use prefixes (di-, tri-)",
59         "Construct IUPAC name (locant-prefix-parent)"
60     ],
61     "numeracySkills": ["Lowest locant determination"],
62     "prerequisiteNodes": [
63         "CHEM_M1_BONDING",
64         "CHEM_M1_HYDROCARBON_BASICS"
65     ],
66     "examQuestionsLinked": ["HSC2022_Q7"],
67     "depthStudyComponents": [],
68     "examples": [
69         "Naming methane, ethane, propane...octane",
70         "Naming 2-methylpropane",
71         "Naming 3-ethylhexane",
72         "Drawing structures from names"
73     ],
74     "media": ["path/to/alkane_naming_tutorial.mp4"]
75 },
76 {
77     "nodeId": "CHM_M7_NOM_N2",
78     "description": "Define and apply IUPAC conventions for naming
79         straight and simple branched alkenes (up to C8), indicating
80         the position of the double bond.",
81     "type": "nomenclature_rule",
82     "bloomTaxonomyLevel": "Apply",
83     "syllabusReference": "CH12-14, M7: Nomenclature (ACSCH127)",
84     "workingScientificallyCategory": "Communicating",
85     "literacySkills": [

```

```

84         "Identify alkene functional group (C=C)",
85         "Identify parent chain containing double bond",
86         "Number parent chain for lowest double bond locant",
87         "Use '-en-' suffix",
88         "Apply alkane substituent rules"
89     ],
90     "numeracySkills": [
91         "Lowest locant determination (priority to double bond)"
92     ],
93     "prerequisiteNodes": [
94         "CHM_M7_NOM_N1",
95         "CHEM_M1_UNSATURATED_HYDROCARBONS"
96     ],
97     "examQuestionsLinked": [],
98     "depthStudyComponents": [],
99     "examples": [
100         "Naming ethene, propene",
101         "Naming but-1-ene vs but-2-ene",
102         "Naming 3-methylpent-1-ene"
103     ],
104     "media": []
105 },
106     .....

```

Listing 1: Excerpt illustrating key elements of the Knowledge Graph definition.

The following listing illustrates the structured format of the knowledge graph, showing:

- The hierarchical organization of the course structure (Chemistry Stage 6 → Module 7 → Organic Chemistry)
- Mapping of content to inquiry questions that drive instruction

- Explicit connections to syllabus outcomes with Bloom's taxonomy levels
- Detailed knowledge node specifications that include:
 - Unique identifiers for each concept/topic
 - Specific syllabi references for traceability
 - Targeted literacy and numeracy skills
 - Prerequisite knowledge connections
 - Associated assessment items and multimedia resources

This structured representation enables systematic instructional design decisions and targeted assessment strategies, allowing for data-driven adaptation of teaching approaches to individual student needs.

Appendix C

Elicit.com AI Research Prompts

To ensure that the content and pedagogical approach were optimised to reflect a research-informed evidence base, the research AI tool from Elicit.com was employed using the following prompts.

Pedagogy for Teaching Organic Chemistry

Prompt: What is an optimal design for a three-lesson teaching sequence (approximately 50-60 minutes per lesson) for NSW Stage 6 Chemistry Module 7 (Organic Reaction Pathways), grounded in cognitive science principles (addressing cognitive load , schema development , visuospatial reasoning , memory, and misconceptions) and active learning methodologies , that effectively facilitates student mastery of functional group interconversions and the construction of multi-step synthesis flowcharts , aligns with syllabus outcomes (CH12-14, CH11/12-6, CH11/12-7) and a specified Educational Knowledge Graph (EduKG), and integrates relevant literacy, numeracy, and ICT skills?

Research Report URL: [Read the report](#)

Google Research Report: Optimizing the Teaching of Organic Reaction Pathways in NSW Stage 6 Chemistry: A Cognitive Science Perspective **Research Report URL:** [Read the report](#)

Google Research Report: Recommending an Optimal d3.js Visualization Method for Organic Chemistry Reaction Pathways **Research Report URL:** [Read the report](#)

Textbook Design Prompt

Prompt: What would be best practice in textbook design for teaching science for school years 7–10 (ages 12–16) that best supports growth in the PISA 2025 Science Framework, particularly for gifted students with ADHD / autism.

Research Report URL: [Read the report](#)

Appendix D
Teaching Resources

Common Resources

MarkSense - Teacher's Guide to Automated Assessment

Introduction

MarkSense streamlines the assessment process, providing near-instantaneous, detailed feedback to your students while significantly reducing your manual marking time. This guide outlines the simple workflow from creating your assessment to delivering graded results.

Workflow Overview

1. **Prepare Assessment:** Create your assessment document and easily generate the corresponding MarkSense template using the built-in tools. Establish the marking rubric for AI assistance.
2. **Distribute:** Give the assessment handout (PDF or print) to students.
3. **Scan:** Students scan their completed handouts using the classroom scanner linked to MarkSense.
4. **Automated Marking:** MarkSense automatically processes each scan: identifies the student and assessment, marks multiple-choice questions, uses AI to analyze written answers against your rubric, generates feedback comments, calculates scores, and compiles a personalized, annotated feedback document for the student.
5. **Deliver Feedback:** The graded feedback document is automatically emailed to each student.
6. **Review (Optional):** Use the MarkSense application to quickly review any flagged items (e.g., answers the AI was unsure about) or browse overall class results.

Step-by-Step Guide for Teachers

1. **Creating Your Assessment:**
 - **Design Handout:** Prepare your assessment document as usual (e.g., in Word, Google Docs, or using Markdown with $\LaTeX$ for scientific content). Ensure clear areas for student responses (MC bubbles, written answer spaces). Export the final handout as a PDF (e.g., `Year7_Science_Term1_Exam.pdf`).
 - **Use MarkSense Setup:** Open the MarkSense application. Select "Create New Assessment Template".
 - Upload your assessment handout PDF (`Year7_Science_Term1_Exam.pdf`).
 - MarkSense will analyze the document. Use the simple on-screen tools to:
 - * Confirm or adjust the areas identified for multiple-choice questions and written answers.
 - * Specify the correct answer for each multiple-choice question (e.g., click the 'C' bubble for Question 1). MarkSense automatically creates the necessary green/red marking guides in the background template.
 - * Define where scores and overall/section comments should appear on the feedback document.
 - * Link or input your marking rubric for each written question to guide the AI analysis.

- Save the template within MarkSense. It will automatically associate this template with the assessment handout.

- **Distribute Handout:** Provide the clean assessment handout PDF (or printouts) to your students. The handout includes an embedded identifier (like a subtle QR code) automatically added by MarkSense during setup, linking it to the template and rubric.

2. Student Completion and Scanning:

- Students complete the handout.
- Students scan their completed handout using the designated MarkSense classroom scanner. The system automatically reads the embedded identifier to know which student and assessment it is, and which template/rubric to use. No manual linking is needed.

3. Automated Marking and Feedback:

- MarkSense processes each scan immediately.
- It marks the MC section, identifying correct/incorrect answers visually on the PDF.
- It analyzes written answers using the AI and your provided rubric, assigning scores and generating initial feedback comments based on the rubric criteria.
- It calculates section and total scores.
- It compiles the annotated PDF with all marks, scores, and comments.
- It automatically emails the completed feedback PDF to the student (using email addresses linked to student IDs in your class lists).

4. Reviewing Results (Optional):

- Open the MarkSense application at any time.
- Select the class and assessment you want to review.
- Browse through the digitally marked student papers.
- MarkSense may flag specific answers (e.g., ambiguous MC marks, low AI confidence on a written answer) for your quick review. You can easily override AI scores or add your own clarifying comments directly in the app.
- View class analytics and generate reports (e.g., CSV scores for upload).

Benefits for You

- **Save Time:** Dramatically reduces time spent on repetitive marking tasks.
- **Focus on Teaching:** Spend more time analyzing results, planning interventions, and interacting with students.
- **Consistent Marking:** Ensures all papers are marked against the same standard using your defined rubric.
- **Faster Feedback:** Students receive detailed feedback almost immediately, reinforcing learning.

MarkSense handles the technical details, allowing you to focus on creating quality assessments and providing timely, meaningful feedback.

Lesson One Resources

Year 12 Chemistry - Lesson Plan 1/3

Mapping the Territory: Visualising Functional Group Interconversions

Mr Haynes

Module 7: Organic Chemistry (Approx. Week 4/5)

Lesson Overview

- **Lesson Title:** Mapping the Territory: Visualising Functional Group Interconversions
- **Duration:** 60 minutes
- **Focus Inquiry Question:** How are different classes of organic compounds interconverted through reaction pathways? (Introduction)
- **Placement:** Assumes students have learned basic nomenclature and reactions connecting Alkanes, Alkenes, Haloalkanes, Alcohols (e.g., Alkene Additions, Alkane/Haloalkane Subs, Alcohol Dehydration/Sub).

Syllabus Alignment & Knowledge Nodes Targeted

- **Outcomes:** CH12-14 (Analyses structure, predicts reactions - single steps), CH11/12-7 (Communicates understanding - using representations)
- **Content:** Implicit review of reactions covered to date. Introduction to representing reaction networks.
- **Knowledge Nodes (Focus on Connections):** CHM_M7_RPROD_N1 (Alkene Add), CHM_M7_RPRO (Alkane Sub), CHM_M7_ALC_N7 (Haloalk->Alc Sub), CHM_M7_ALC_N5 (Alc->Haloalk Sub), CHM_M7_ALC_N4 (Alc Dehyd).

Student Learning Objectives (Aligned with Nodes & Cognitive Strategies)

Students will be able to:

- Explain the concept of a "reaction map" for organic chemistry. [Understand]
- Identify key functional groups (nodes) and known reaction pathways (chords/links) using the chord-diagram visualisation tool. [Apply - S2 Visualisation]
- Use the tool's features (hover, filter) to retrieve information about specific known reactions (type, reagents/conditions). [Apply - S2]
- Relate the visual representation on the map to symbolic reaction equations learned previously. [Analyse - S2 Dual Coding]
- Begin constructing a mental schema of interconnected reactions. [Understand - S3 Concept Mapping Intro]

Literacy Use terminology for functional groups and basic reaction types correctly.

Lesson Structure & Activities

Introduction (10 mins)

- **Teacher Activity:** Display Inquiry Question. Quick recap quiz (retrieval practice): "Name the product when HBr adds to propene." "What condition is needed to substitute Cl onto methane?". Introduce the idea that reactions form a network/map, not just a list. State lesson objective: Learning to use a visual tool to explore this map.
- **Student Activity:** Answer recap questions. Listen to introduction.
- **Pedagogy Focus:** Activate Prior Knowledge, Set Context, Retrieval Practice (supports S4).

Exploration: Introducing the Visualisation Tool (25 mins)

- **Teacher Activity:** Introduce and demonstrate the interactive Chord Diagram visualisation tool (projected). Explain: Nodes = Functional Groups, Chords = Reactions. Show how chord colour relates to reaction type (using legend/checkboxes). Demonstrate hover-over feature for reaction details (reagents/conditions). Focus **only** on nodes/reactions students have already learned (e.g., Alkene, Alcohol, Haloalkane connections). Explicitly link tool view to a written equation shown previously. [S1 Explicit Instruction, S2 Visualisation]
- **Student Activity:** Observe demonstration. Ask clarifying questions about the tool.
- **Pedagogy Focus:** Tool Introduction, Linking Visual Representation to Symbolic (Dual Coding S2), Managing Cognitive Load (showing only known parts first).
- **ICT Integration:** Chord Diagram Tool, Projector.

Guided Exploration & Consolidation (25 mins)

- **Teacher Activity:** Distribute Worksheet 1 and provide link/access to the visualisation tool. Guide students (working individually or pairs on devices) through Worksheet 1 tasks. Circulate, check understanding, and assist with tool usage. Lead brief class discussion on findings (e.g., "What reaction types connect Alcohols and Alkenes according to the map?"). Consolidate the idea of the map as a growing organiser for their knowledge. Assign Exit Ticket. [S3 Concept Mapping Intro, S2 Visualisation]
- **Student Activity:** Use the visualisation tool on devices to complete Worksheet 1 tasks (e.g., identifying connections, retrieving reagent info via hover, filtering). Participate in discussion. Complete Exit Ticket.
- **Pedagogy Focus:** Active Learning, Guided Inquiry using Visualisation Tool, Reinforcing Connections, Formative Assessment.
- **ICT Integration:** Chord Diagram Tool (Student Use), Devices.

Resources Required

- Interactive Chord Diagram Visualisation Tool.
- Projector.
- Student devices with internet access (recommended 1:1 or 1:2).
- Worksheet 1 (separate file - see below).
- Exit Ticket questions (prepared separately, e.g., "Using the visualizer, what reagent is shown for converting an Alkene to an Alcohol?").

Assessment

- **Formative:** Teacher observation of student engagement and tool use. Responses during class discussion. Review of Worksheet 1 answers. Analysis of Exit Ticket responses (checking basic tool interpretation).

Differentiation

- **Support:** Provide a simplified version of Worksheet 1 with more direct prompts. Pair students for tool exploration. Pre-fill parts of the map diagram on the worksheet.
- **Extension:** Ask students to predict what other connections might exist based on functional group similarities. Challenge students to find a reaction on the map they haven't learned yet and hypothesise about its type.

Lesson 1: Mapping the Territory

Visualising Functional Group Interconversions

Mr Haynes

Gosford High School

Module 7: Organic Chemistry

Outline

- 1 Introduction
- 2 The Visualisation Tool
- 3 Guided Exploration
- 4 Summary

Introduction: Why Map Reactions?

Focus Inquiry Question: How are different classes of organic compounds interconverted through reaction pathways?

- **Recap Quiz:** (Teacher to ask 1-2 quick questions on recent reactions, e.g., alkene addition product, alkane substitution condition).
- Organic reactions don't exist in isolation – they form an **interconnected network**.
- Learning the connections is key to understanding organic chemistry and planning syntheses.
- **Today's Goal:** Learn to use a visual tool (Chord Diagram) to explore this reaction network based on what we've learned so far.

Understanding the Visualisation

This tool helps us see the "map" of organic reactions.

Key Features:

- **Nodes** (Outer Segments): Represent Functional Groups (e.g., Alkane, Alkene, Alcohol).
- **Chords** (Inner Bands): Represent Reactions linking functional groups.
- **Colour Coding**: Indicates Reaction Type (Check legend/checkboxes - e.g., Addition, Substitution).
- **Hover/Click**: Reveals details for a specific reaction (Reagents, Conditions).
- **Filtering**: Checkboxes allow focusing on specific reaction types.

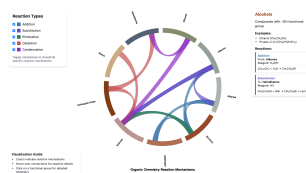


Figure: Chord Diagram Example

Using the Tool for Known Reactions

Now, let's use the tool to map out the reactions we've already covered.

Activity Prep (Refer to Activity Sheet 1 / Worksheet 1):

- Access the Chord Diagram tool on your device.
- **Your Task:** Use the tool to find the connections between Alkanes, Alkenes, Haloalkanes, and Alcohols.
- For each connection (reaction):
 - Identify the reaction type (using colour/filter).
 - Find the reagents/conditions (using hover).
 - Record your findings on Worksheet 1.
- **Focus:** Connecting the visual map to the reactions you already know.

(Teacher circulates to assist with tool usage and understanding)

Mapping the Territory: Key Takeaways

- Organic reactions form an interconnected network or "map".
- The Chord Diagram tool helps visualise this network:
 - Nodes = Functional Groups
 - Chords = Reactions (coloured by type)
 - Hover = Reagents/Conditions
- This tool helps organise our knowledge and see the bigger picture.

Next Steps:

- Complete Worksheet 1.
- Complete Exit Ticket (check tool interpretation).
- **Preview Lesson 2:** Using the map to plan short synthesis pathways (connecting reactions).

Thank you!

Questions?

Year 12 Chemistry - Activity Sheet 1

Guided Exploration: Reaction Visualisation Tool

Organic Chemistry

Module 7 - Lesson 1

Aim

To familiarise yourself with the interactive Chord Diagram Visualisation tool and use it to identify known organic reaction pathways and their key details.

Tool Access

Use the link or application provided by your teacher to access the Chord Diagram tool on your device.

Part A: Tool Familiarisation

- Identify the main components: Outer segments (Nodes), Inner bands (Chords/Links).
- Locate the legend or checkboxes that explain the colour-coding for reaction types (e.g., Addition, Substitution, Elimination, Oxidation, Condensation).
- Practice hovering your mouse cursor over different Nodes and Chords. Observe what information appears (e.g., Functional group name, Reaction details like reagents/conditions).
- Practice using the checkboxes (if available) to filter the view and show only certain reaction types.

Part B: Guided Exploration (Complete relevant sections on Worksheet 1)

Task 1: Focus on Alkenes

1. Click on or interact with the 'Alkene' node.
2. Observe the chords connecting to it.
3. For the connection representing the reaction of an Alkene to form an Alcohol:
 - Identify the reaction type using the colour code / filter.
 - Hover to find the specific reagents/conditions (e.g., Hydration conditions). Record these on Worksheet 1.
4. Repeat for the connection representing the reaction of an Alkene to form a Haloalkane (via HX addition). Record the reaction type and typical reagent (e.g., HBr) on Worksheet 1.
5. Repeat for the connection representing the reaction of an Alkene to form a Haloalkane (via X₂ addition). Record the reaction type and typical reagent (e.g., Br₂).

Task 2: Focus on Alcohols

1. Click on or interact with the 'Alcohol' node.
2. Find the chord representing the conversion of an Alcohol back to an Alkene.
 - Identify the reaction type (Dehydration/Elimination).
 - Hover to find the specific reagents/conditions. Record these on Worksheet 1.
3. Find the chord representing the conversion of an Alcohol to a Haloalkane.
 - Identify the reaction type (Substitution).
 - Hover to find the typical reagent. Record this on Worksheet 1.

Task 3: Focus on Haloalkanes

1. Click on or interact with the 'Haloalkane' node.
2. Find the chord representing the conversion of a Haloalkane to an Alcohol.
 - Identify the reaction type (Substitution).
 - Hover to find the typical reagent/conditions. Record these on Worksheet 1.
3. Does the map (based on reactions learned so far) show a direct conversion from an Alkane to a Haloalkane? Identify the reaction type and condition required.

Task 4: Synthesise (Worksheet 1, Q6) Use your findings to draw the connections between the four main functional groups studied so far (Alkanes, Alkenes, Haloalkanes, Alcohols) on Worksheet 1.

Year 12 Chemistry - Worksheet 1

Mapping the Territory: Visualising Reactions

Student Name: _____ ID: _____

Module 7 - Lesson 1

Part 1: The Reaction Map Concept

1. In your own words, what is a "reaction map" in organic chemistry? Why might it be useful?

2. The chord diagram tool uses nodes and chords/links. What does each represent?

- Nodes (Outer segments): _____
- Chords / Links (Inner connecting bands): _____

Part 2: Exploring Known Connections with the Tool

Instructions: Use the interactive Chord Diagram Visualisation tool provided by your teacher to answer the following questions. Focus on the reactions you have already learned (connecting Alkanes, Alkenes, Haloalkanes, Alcohols).

3. Find the node for **Alkenes**.

- List the functional groups that Alkenes are shown to be directly connected to via reactions *you have learned so far*.
- Select one of these connections (e.g., Alkene $\rightarrow$ Alcohol). What is the reaction type indicated by the chord colour/filter?
- Hover over this specific chord/link. What reagent(s) and/or conditions are displayed for this transformation (e.g., $\text{H}_2\text{O} / \text{H}^+$)?

4. Find the node for **Alcohols**.

- According to the map (based on reactions learned), can an Alcohol be directly converted back to an Alkene? (Yes/No)
- If yes, what is the reaction type and the key reagent/condition shown when you hover over that link?

- c) According to the map (based on reactions learned), can an Alcohol be directly converted to a Haloalkane? (Yes/No)
- d) If yes, what reagent is shown for this transformation?

5. Find the connection between **Haloalkanes** and **Alcohols**.

- a) What reagent is needed to convert a Haloalkane to an Alcohol according to the tool?
- b) What type of reaction is this (use colour/filter)?

6. Based on your exploration, draw a simple diagram below showing only the nodes for Alkane, Alkene, Haloalkane, and Alcohol, and draw arrows representing the direct, one-step reactions you have identified between them using the tool. Label each arrow with the reaction type (e.g., Addition, Substitution, Dehydration). [S3 Concept Mapping]

Lesson Two Resources

Year 12 Chemistry - Lesson Plan 2/3

Navigating Pathways: Connecting Reactions for Short Syntheses

Mr Haynes

Module 7: Organic Chemistry (Approx. Week 6/7)

Lesson Overview

- **Lesson Title:** Navigating Pathways: Connecting Reactions for Short Syntheses
- **Duration:** 60 minutes
- **Focus Inquiry Question:** How are different classes of organic compounds interconverted through reaction pathways? (Focus on 2-3 step sequences)
- **Placement:** Assumes L1 completed. Assumes students have now learned Alcohol Oxidation (CHM_M7_ALC_N6) and Esterification (CHM_M7_ESTER_N1, CHM_M7_NOM_N11).

Syllabus Alignment & Knowledge Nodes Targeted

- **Outcomes:** CH12-14 (Predicts reactions involving carbon compounds), CH11/12-6 (Solves scientific problems - synthesis planning), CH11/12-7 (Communicates understanding).
- **Content:** Applying knowledge of multiple reaction steps in sequence. Introduction to synthesis planning logic.
- **Knowledge Nodes (Focus):** CHM_M7_ALC_N6 (Oxidation), CHM_M7_ESTER_N1 (Esterification), CHM_M7_NOM_N11 (Ester Naming). Review/Application of nodes from L1.

Student Learning Objectives (Aligned with Nodes & Cognitive Strategies)

Students will be able to:

- Identify multi-step pathways (2-3 steps) between functional groups using the chord-diagram tool. [Apply - S2 Visualisation]
- Propose a logical sequence of known reactions to achieve a simple synthetic transformation. [Analyse - CH11/12-6]
- Identify necessary reagents and conditions for each step in a proposed short synthesis. [Apply - CH12-14]
- Name reactants, intermediates, and products (including simple esters) in a short synthesis pathway. [Apply - CHM_M7_NOM_N11]
- Articulate the planning process for solving a simple synthesis problem. [Apply - S6 Metacognition]
- Update and utilise their reaction map (mental or visual) incorporating oxidation and esterification. [Apply - S3 Concept Mapping]

Literacy Describe a short synthesis pathway using correct terminology and reaction representation.

Lesson Structure & Activities

Introduction & Retrieval Practice (10 mins)

- **Teacher Activity:** Brief retrieval quiz: "Show the reagents needed to convert 1-chloropropane to propan-1-ol." "What functional group results from dehydrating an alcohol?". Briefly review the chord diagram from L1, highlighting the previously learned connections. Add new nodes/reactions (Oxidation, Esterification) to the discussion/visualisation. [S3 Update Map, supports S4]
- **Student Activity:** Answer retrieval questions. Observe map update.
- **Pedagogy Focus:** Retrieval Practice, Activation of Prior Knowledge, Integrating New Knowledge into Schema (S3).

Modelling Short Synthesis Planning (15 mins)

- **Teacher Activity:** Explicitly model solving a 2-step synthesis problem (e.g., Ethene → Ethanoic Acid).
 - Step 1: Identify Start/End functional groups.
 - Step 2: Use the chord diagram tool to find path(s) - "Ethene to Alcohol (Addition), then Alcohol to Carboxylic Acid (Oxidation)". Identify intermediate (Ethanol).
 - Step 3: Write out the sequence, adding specific reagents/conditions for each step learned previously (hover on tool or recall).
 - Step 4: "Think aloud" the metacognitive process: "I need to get from alkene to acid. The map shows I can go via alcohol. First step is hydration... second step is oxidation of a primary alcohol..." [S1 Explicit Instruction, S2 Visualisation, S6 Metacognition]
- **Student Activity:** Follow the modelled example. Ask clarifying questions.
- **Pedagogy Focus:** Modelling Problem-Solving Process, Explicit use of Visualisation Tool for Planning, Metacognitive Scaffolding (S6).

Paired Problem Solving (25 mins)

- **Teacher Activity:** Provide pairs of students with simple 2 or 3-step synthesis problems (via Activity Sheet 2 or whiteboard). Encourage them to use the chord diagram tool *first* to plan their route, then write down the detailed steps (structures/names, reagents/conditions) on mini-whiteboards or paper. Circulate, prompt with planning questions ("What intermediate is needed?", "What reaction achieves that?", "Check the tool for reagents."). [S4 Interleaving if problems mix reaction types]
- **Student Activity:** Work in pairs. Use the visualisation tool to plan pathways for given synthesis problems. Draft the reaction sequence with structures, names, reagents, conditions. Discuss strategy with partner.
- **Pedagogy Focus:** Collaborative Learning, Applying Knowledge, Problem Solving (CH11/12-6), Active use of Visualisation Tool, Metacognitive Practice (S6 prompts).
- **ICT Integration:** Chord Diagram Tool (Student Use), Devices.

Sharing Consolidation (10 mins)

- **Teacher Activity:** Select 1-2 pairs to present their pathway for one problem on the board. Facilitate discussion – "Did anyone find a different route using the tool?". Briefly summarise the process of using the map for planning.

- **Student Activity:** Share proposed pathway if selected. Observe and comment on others' pathways. Ask questions.
- **Pedagogy Focus:** Communicating Solutions (CH11/12-7), Peer Learning, Consolidating the Planning Strategy.

Resources Required

- Interactive Chord Diagram Visualisation Tool.
- Projector / Whiteboard.
- Student devices with internet access.
- Mini-whiteboards or paper for drafting pathways.
- Activity Sheet 2 (containing 2/3-step synthesis problems - see below).

Assessment

- **Formative:** Observation of paired problem-solving process (use of tool, discussion, logic). Review of drafted pathways on mini-whiteboards/paper (checking for correct intermediates, reagents, conditions). Quality of participation in sharing session.

Differentiation

- **Support:** Provide partially completed pathways (e.g., give the intermediate). Offer a list of possible reagents to choose from. Pair students strategically.
- **Extension:** Challenge students to find the *shortest* possible route if multiple exist. Ask them to propose a synthesis for a slightly more complex target requiring 3 steps.

Lesson 2: Navigating Pathways

Connecting Reactions for Short Syntheses

Mr Haynes

Gosford High School

Module 7: Organic Chemistry

Outline

- 1 Recap & New Connections
- 2 Planning Short Syntheses
- 3 Paired Problem Solving
- 4 Summary

Recap & Expanding the Map

Retrieval Practice: (Teacher asks questions, e.g., "Reagent for Haloalkane $\rightarrow$ Alcohol?", "Product of alcohol dehydration?") **Review:**

Briefly show Chord Diagram from L1. Remind students of the connections learned. **New Reactions (Adding to our map):**

- **Alcohol Oxidation [CHM_M7_ALC_N6]:**
 - Primary Alcohol $\xrightarrow{[O]}$ Aldehyde $\xrightarrow{[O]}$ Carboxylic Acid
 - Secondary Alcohol $\xrightarrow{[O]}$ Ketone
 - Tertiary Alcohol $\xrightarrow{[O]}$ No reaction (usually)
 - (*Show these connections on the projected visualiser*)
- **Esterification [CHM_M7_ESTER_N1]:**
 - Carboxylic Acid + Alcohol $\xrightarrow{H^+/\Delta}$ Ester + H₂O
 - (*Show this connection on the visualiser*)
- **Ester Naming [CHM_M7_NOM_N11]:** (Brief mention - e.g., *Alkyl Alkanoate*)

From A to C via B: Planning Pathways

Often, we can't get from starting material (A) to target product (C) in one step. We need intermediate compounds (B). **The Strategy:**

- 1 Identify Starting and Target Functional Groups.
- 2 **Use the Chord Diagram Tool:** Find pathway(s) connecting Start to Target. Identify the intermediate functional group(s).
- 3 Write out the sequence of reactions.
- 4 Add specific reagents and conditions for each step.

Modelling: Ethene to Ethanoic Acid

Problem: Convert Ethene (C_2H_4) to Ethanoic Acid (CH_3COOH).

Teacher Modelling ("Think Aloud" - S6):

- 1 "Start is Alkene, Target is Carboxylic Acid."
- 2 "Look at the map (Chord Diagram)... Alkene connects to Alcohol. Alcohol connects to Aldehyde, which connects to Carboxylic Acid (for primary). So, the path is Alkene $\rightarrow$ Alcohol $\rightarrow$ Carboxylic Acid. Intermediate = Alcohol (Ethanol)."
- 3 "Write the sequence:"
 - Step 1 (Alkene $\rightarrow$ Alcohol): $\text{CH}_2=\text{CH}_2 \longrightarrow \text{CH}_3\text{CH}_2\text{OH}$. "What reagent? Map hover/recall... Hydration." Reagent: $\text{H}_2\text{O} / \text{H}^+$.
 - Step 2 (Alcohol $\rightarrow$ Acid): $\text{CH}_3\text{CH}_2\text{OH} \longrightarrow \text{CH}_3\text{COOH}$. "What reagent? Map hover/recall... Oxidation of primary alcohol." Reagent: Strong Oxidising Agent (e.g., $\text{Cr}_2\text{O}_7^{2-} / \text{H}^+$), shown as [O].
- 4 "Check: Path makes sense, reagents identified."

(Show final written sequence clearly)



Paired Problem Solving Activity

Now, work with your partner on the problems from Activity Sheet 2.

Instructions Recap:

- Use the Chord Diagram tool to **PLAN** your route first.
- Write down the detailed steps on mini-whiteboard/paper/Worksheet 2:
 - Structures (or names).
 - Reagents & Conditions for each arrow.
- Discuss your strategy with your partner. Use the metacognitive prompts!

(Teacher circulates, assists, and prompts using S6 questions)

Navigating Pathways: Key Takeaways

- We expanded our reaction map with Alcohol Oxidation and Esterification.
- We practiced using the map (visualiser) to plan short (2-3 step) synthesis pathways.
- The planning process: Identify Start/Target → Find Route via Intermediates (using map) → Add Reagents/Conditions.
- This moves us from knowing single reactions to connecting them.

Next Steps:

- Review drafted pathways.
- **Preview Lesson 3:** Tackling more complex synthesis problems and learning to communicate them using formal flowcharts (Syllabus requirement!).

Thank you!

Questions?

Year 12 Chemistry - Activity Sheet 2

Paired Problem Solving: Short Syntheses

Organic Chemistry

Module 7 - Lesson 2

Aim

To practice planning and representing short (2-3 step) organic synthesis pathways using your knowledge of reactions and the visualisation tool.

Instructions

Work with your partner. For each problem below:

1. Identify the starting material and the target product.
2. Use the Chord Diagram Visualisation tool to help you brainstorm possible reaction sequences. Identify necessary intermediates.
3. On your mini-whiteboard or paper (or Worksheet 2), write out the step-by-step reaction pathway.
4. For each step, clearly show the structure (or IUPAC name) of the reactant and product.
5. Above/below the reaction arrow for each step, write the necessary reagent(s) and conditions.
6. Be prepared to explain your chosen pathway.

Synthesis Problems

Problem Set A: Propose a 2-step synthesis pathway for the conversion of **ethene** to **ethanoic acid**.

Problem Set B: Propose a synthesis pathway for the conversion of **propane** to **propanone**. (Hint: This might take more than 2 steps).

Problem Set C: Propose a synthesis pathway for the conversion of **1-bromobutane** to **butanoic acid**.

Problem Set D (Ester Challenge): Propose a synthesis pathway for the conversion of **propene** to **propyl ethanoate**. (Hint: You'll need to synthesise both the alcohol and the acid parts, potentially from the propene starting material if possible, or assume ethanoic acid is available).

Metacognitive Prompts (Consider while working):

- What functional group change is needed?
- What reaction(s) can achieve this change? (Check the visualizer!)
- What reagents/conditions are specific to that reaction?
- Is there another way to make this connection?

Year 12 Chemistry - Worksheet 2

Navigating Pathways: Short Syntheses

Student Name: _____ ID: _____

Module 7 - Lesson 2

Part 1: New Reactions - Oxidation & Esterification

1. Oxidation of Alcohols (Node: CHM_M7_ALC_N6):

- a) What type of alcohol (primary, secondary, tertiary) is oxidised to form an aldehyde (which can be further oxidised to a carboxylic acid)?
- b) What type of alcohol is oxidised to form a ketone?
- c) What type of alcohol is generally resistant to oxidation under these conditions?
- d) Name a common oxidising agent used for these reactions (often represented as [O]).

2. Esterification (Node: CHM_M7_ESTER_N1):

- a) What two types of functional groups react to form an ester?
- b) What catalyst and condition are typically required for esterification?
- c) What small molecule is also produced during esterification?

3. Ester Naming (Node: CHM_M7_NOM_N11): Name the following esters:

- a) The ester formed from methanol and ethanoic acid: _____
- b) The ester formed from propan-1-ol and propanoic acid: _____
- c) Draw the structure of ethyl propanoate:

Part 2: Planning Short Syntheses

Instructions: For the synthesis problems assigned in class (see Activity Sheet 2), use the space below to plan your pathway. Use the Chord Diagram tool to help identify intermediates and reaction types.

For each step, show: Reactant Structure/Name $\xrightarrow{\text{Reagent(s)/Conditions}}$ Product Structure/Name.

Problem 1: [e.g., Ethene to Ethanoic Acid]

Planning Notes (Intermediates? Reaction Types?):

Pathway: Step 1: Step 2:

Problem 2: [e.g., Propane to Propanone]

Planning Notes:

Pathway: Step 1: Step 2: (Add more steps if needed)

Problem 3: [e.g., 1-Bromobutane to Butanoic Acid]

Planning Notes:

Pathway: Step 1: Step 2: (Add more steps if needed)

Lesson Three Resources

Year 12 Chemistry - Lesson Plan 3/3

Designing & Communicating Syntheses: Mastering Flowcharts

Mr Haynes

Module 7: Organic Chemistry (Approx. Week 9/10)

Lesson Overview

- **Lesson Title:** Designing & Communicating Syntheses: Mastering Flowcharts
- **Duration:** 60 minutes
- **Focus Inquiry Question:** How can we represent multi-step organic syntheses? (Culminating task)
- **Placement:** Assumes L1 & L2 completed. Assumes all relevant reactions from EduKG subset (Alkene Add/Sub, Haloalkane Sub, Alcohol Sub/Dehyd/Oxid, Esterification) have been covered in the module.

Syllabus Alignment & Knowledge Nodes Targeted

- **Outcomes:** CH12-14 (Analyses structure, predicts reactions), CH11/12-6 (Solves scientific problems - complex synthesis), CH11/12-7 (Communicates scientific understanding - flowchart conventions).
- **Content:** Constructing multi-step synthesis flowcharts with correct conventions, reagents, and conditions. Applying integrated knowledge of reaction pathways.
- **Knowledge Nodes (Focus):** CHM_M7_SYNTH_N1 (Draft and construct flow charts...). Application of all prerequisite reaction nodes.

Student Learning Objectives (Aligned with Nodes & Cognitive Strategies)

Students will be able to:

- Analyse a multi-step synthesis problem to identify starting material, target product, and required functional group changes. [Analyse - CH11/12-6]
- Utilise the chord-diagram tool and reaction knowledge to devise a logical multi-step synthesis pathway. [Create - CH11/12-6]
- Apply standard flowchart conventions to represent a multi-step synthesis accurately. [Apply - CHM_M7_SYNTH_N1, CH11/12-7]
- Include correct structural formulae (or IUPAC names) for all intermediates and products in a flowchart. [Apply - CH11/12-7]
- Specify correct reagents and conditions for each step in the flowchart. [Apply - CH12-14]
- Justify the chosen reaction pathway and steps. [Evaluate - S6 Metacognition]

Literacy Communicate a complex chemical process clearly and conventionally using a flowchart.

Numeracy Implicitly check atom conservation through balanced steps/structures.

Lesson Structure & Activities

Introduction & Map Review (10 mins)

- **Teacher Activity:** Briefly review the "complete" reaction map using the chord diagram, emphasizing the network of possibilities learned over the module. Pose a quick challenge question: "Can we directly convert an alkane to an ester using the reactions we've learned? Why/why not? What intermediate(s) would be needed?". State lesson objective: To plan complex syntheses and communicate them using formal flowcharts. [S3 Map Review]
- **Student Activity:** Participate in map review and challenge question discussion.
- **Pedagogy Focus:** Consolidation of Schema (S3), Assessing Integrated Understanding.

Instruction & Modelling: Flowchart Construction (15 mins)

- **Teacher Activity:** Explicitly teach standard flowchart conventions for chemical synthesis (referencing CHM_M7_SYNTH_N1 literacy skills): Boxes for compounds (structure or name), Arrows for reactions, Reagents/conditions written above/below arrows. Model solving a more complex (e.g., 3-step) synthesis problem (e.g., Propane $\rightarrow$ Propan-1-ol $\rightarrow$ Propanal $\rightarrow$ Propanoic Acid).
 - First, use the chord diagram tool to plan the route, "thinking aloud" the choices.
 - Then, translate the planned route step-by-step into a correctly formatted flowchart on the board/projector. Emphasise clarity and inclusion of all required details. [S1 Explicit Instruction, S2 Visualisation, S6 Metacognition]
- **Student Activity:** Take notes on flowchart conventions (can be added to Worksheet 3). Follow the modelled example. Ask questions about conventions or the synthesis logic.
- **Pedagogy Focus:** Explicit Teaching of Communication Conventions (CH11/12-7), Modelling the Link between Planning (Tool) and Communicating (Flowchart).

Group Synthesis Challenge (25 mins)

- **Teacher Activity:** Divide students into small groups (3-4). Provide each group with a different, challenging synthesis problem (e.g., starting from a simple alkane/alkene, synthesise a specific ester or ketone) via Activity Sheet 3. Provide access to the chord diagram tool for planning and materials for flowchart construction (large paper, mini-whiteboards, or digital tool). Instruct groups to produce a complete, conventional flowchart as their solution. Circulate to facilitate, guide, and ask probing questions (S6 prompts). [Problem-Based Learning element, S4 Interleaving via varied problems]
- **Student Activity:** Work collaboratively in groups. Analyse the assigned problem. Use the chord diagram tool to brainstorm and plan a viable pathway. Construct the synthesis flowchart, ensuring correct structures/names, reagents, conditions, and conventions. Discuss and justify steps within the group.
- **Pedagogy Focus:** Collaborative Problem Solving (CH11/12-6), Application of Integrated Knowledge (Create Level), Practice with Communication Conventions (CH11/12-7), Active Learning.
- **ICT Integration:** Chord Diagram Tool (Student Use), Optional Digital Whiteboard/Drawing Tool.

Peer Review / Gallery Walk (10 mins)

- **Teacher Activity:** Facilitate a gallery walk or structured peer review. Groups display their flowcharts. Provide a simple peer review checklist (on Activity Sheet 3 or separate) focusing on: Logical sequence, Correct structures/names, Correct reagents/conditions, Clear conventions. Encourage constructive feedback. Briefly summarise common successes/errors observed. Assign Exit Ticket.
- **Student Activity:** Display group flowchart. View and provide feedback on other groups' flowcharts using the checklist. Discuss feedback received. Complete Exit Ticket.
- **Pedagogy Focus:** Peer Assessment, Reinforcing Communication Standards (CH11/12-7), Identifying Common Errors, Formative Assessment.

Resources Required

- Interactive Chord Diagram Visualisation Tool.
- Projector / Whiteboard.
- Student devices (for tool access).
- Materials for flowchart construction (large paper/posters, markers, mini-whiteboards, or digital collaborative whiteboard).
- Activity Sheet 3 (containing synthesis challenge problems and peer review checklist - see below).
- Worksheet 3 (summary of flowchart conventions - see below).
- Exit Ticket question (prepared separately, e.g., "Outline the first two steps (reactants, reagents) needed to convert but-1-ene into ethyl ethanoate.").

Assessment

- **Formative:** Observation of group collaboration, planning strategies, and flowchart construction. Evaluation of final group flowcharts based on logic, accuracy, detail, and conventions. Quality of peer feedback provided. Analysis of Exit Ticket responses.

Differentiation

- **Support:** Assign specific roles within groups (e.g., planner, recorder, reagent checker). Provide partially solved synthesis problems or flowcharts with gaps to fill. Offer a checklist of key reactions.
- **Extension:** Challenge groups to find the most efficient (fewest steps) pathway. Ask them to propose a synthesis for a target molecule not directly derivable using only the learned reactions, requiring slight modification or an additional step. Ask groups to include potential side reactions or yield considerations (conceptual).

Lesson 3: Designing & Communicating Syntheses

Mastering Flowcharts (CHM_M7_SYNTH_N1)

Mr Haynes

Gosford High School

Module 7: Organic Chemistry

Outline

- 1 The Complete Picture
- 2 Flowchart Conventions
- 3 Modelling Practice
- 4 Review Summary

The Interconnected Network

Let's look at the full reaction map (Chord Diagram) incorporating all the reactions we've learned in this module.

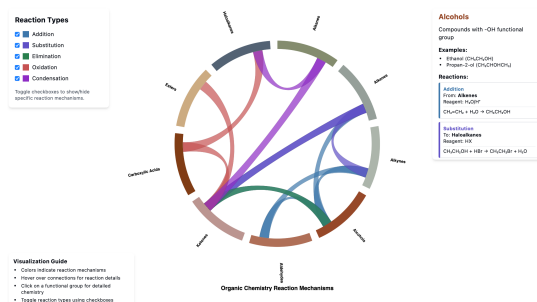


Figure: Overview of Key Functional Group Interconversions

Challenge Question: Can we directly convert an Alkane to an Ester using

P. Haynes (GHS)

Org Chem: Designing Syntheses

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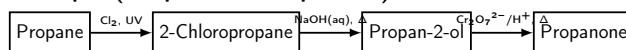
Communicating Synthesis Clearly (CHM_M7_SYNTH_N1)

Flowcharts are the standard way to show multi-step syntheses. **Key**

Conventions (Worksheet 3):

- **Compounds:** In boxes (Use IUPAC Name or Structural Formula).
- **Reactions:** Use arrows $\rightarrow$.
- **Reagents/Conditions:** Write above/below the arrow for THAT step.
- **Layout:** Logical flow (e.g., top-to-bottom).

Example (Propane $\rightarrow$ Propanone):



Plan then Draw: Example

Problem: Synthesise Propanoic Acid from Propane. **Teacher Modelling**

("Think Aloud"):

- 1 "Start=Alkane, Target=Carboxylic Acid."
- 2 "Map Check: Alkane $\rightarrow$ Haloalkane $\rightarrow$ Alcohol (Primary needed for Acid) $\rightarrow$ Carboxylic Acid. Path looks viable." Intermediates: 1-Chloropropane, Propan-1-ol.
- 3 "Translate to Flowchart:" (Draw step-by-step on board/slide)
 - Box 1: Propane ($\text{CH}_3\text{CH}_2\text{CH}_3$)
 - Arrow 1: Reagents Cl_2 , UV Light (Free radical sub gives mix, but targets pathway) $\rightarrow$ Box 2: 1-Chloropropane ($\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$)
 - Arrow 2: Reagents $\text{NaOH}(\text{aq})$, heat (Substitution) $\rightarrow$ Box 3: Propan-1-ol ($\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$)
 - Arrow 3: Reagents $\text{Cr}_2\text{O}_7^{2-}/\text{H}^+$, heat (Oxidation [O]) $\rightarrow$ Box 4: Propanoic Acid ($\text{CH}_3\text{CH}_2\text{COOH}$)
- 4 "Ensure all reagents/conditions and structures/names are correct."

Design Build Your Flowchart

Work in your groups on the challenge problem from Activity Sheet 3. **Your**

Task Recap:

- Analyse the problem (Start/Target).
- Plan your route using the Chord Diagram tool.
- Construct a formal, accurate flowchart on your paper/board.
- Include all compounds (structures/names) and reagents/conditions.

Goal: Create a chemically correct and clearly communicated synthesis plan. (*Teacher circulates, facilitates group work, asks probing questions*)

Evaluating Flowcharts

Time for Peer Review / Gallery Walk.

- Display your group's flowchart.
- Use the checklist on Activity Sheet 3 to review another group's work.
- Focus on: Logical Steps, Correct Chemistry (Structures/Reagents), Clear Conventions.
- Provide constructive feedback.

Final Summary:

- Synthesis requires integrated knowledge and planning (use the map!).
- Flowcharts are the standard way to communicate these plans accurately (CH11/12-7).

Next Steps: Complete Exit Ticket. Revise reaction pathways for upcoming assessments.

Thank you!

Questions?

Year 12 Chemistry - Activity Sheet 3

Group Synthesis Challenge & Flowchart Peer Review

Organic Chemistry

Module 7 - Lesson 3

Aim

To apply your integrated knowledge of organic reactions to design a multi-step synthesis pathway and communicate it using a conventional flowchart. To provide constructive feedback on peers' flowcharts.

Part A: Group Synthesis Challenge

Instructions: Work in your assigned group. Your task is to design a synthesis pathway for the problem below and represent it as a detailed, accurate flowchart.

1. Analyse the starting material and target product.
2. Use the Chord Diagram Visualisation tool and your knowledge to plan a logical sequence of reactions.
3. Construct a flowchart on the provided materials (paper/whiteboard/digital).
4. Ensure your flowchart includes:
 - Correct structures or IUPAC names for all compounds (starting material, intermediates, product) in boxes.
 - Arrows clearly indicating each reaction step.
 - Accurate reagents and conditions listed for every step on the arrows.
 - Adherence to standard flowchart conventions.
5. Be prepared to justify your chosen pathway.

Challenge Problem (Teacher will assign one per group)

Problem 1: Design a synthesis pathway to produce **ethyl propanoate** starting from **ethene** and **propane**. (Assume necessary inorganic reagents are available).

Problem 2: Design a synthesis pathway to produce **butanone** starting from **but-1-ene**.

Problem 3: Design a synthesis pathway to produce **1,2-dichloroethane** starting from **ethanol**.

Problem 4: Design a synthesis pathway to produce **propanoic acid** starting from **propane**.

Part B: Flowchart Peer Review Checklist

Instructions: When reviewing another group's flowchart, consider the following criteria:

Criteria	Description	Check ()
1. Logical Pathway	Does the sequence of reactions make chemical sense to get from start to target?	
2. Correct Structures/Names	Are the structures or names shown for reactants, intermediates, and products accurate?	
3. Correct Reagents/Conditions	Are the specified reagents and conditions appropriate for each reaction step shown?	
4. Flowchart Conventions	Are compounds in boxes? Are arrows used correctly? Are reagents/conditions placed appropriately on arrows?	
5. Clarity & Neatness	Is the flowchart easy to read and understand?	

Constructive Feedback / Comments:

Year 12 Chemistry - Worksheet 3

Designing Syntheses: Flowchart Conventions

Student Name: _____ ID: _____

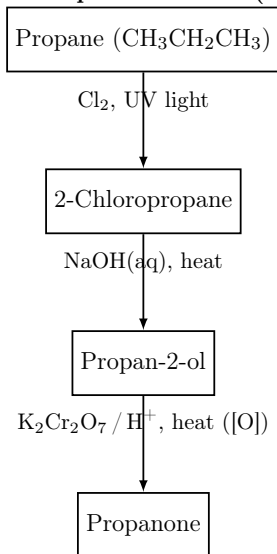
Module 7 - Lesson 3

Part 1: Flowchart Conventions (Node: CHM_M7_SYNTH_N1)

Flowcharts are used to communicate multi-step synthesis pathways clearly and conventionally. Key conventions include:

- **Compounds:** Represented inside boxes. Use either the correct IUPAC name or the structural formula. Be consistent.
- **Reactions:** Represented by arrows pointing from reactant(s) to product(s).
- **Reagents & Conditions:** Written clearly above or below the reaction arrow corresponding to that specific step. Include necessary catalysts, solvents, heat (Δ), UV light, etc.
- **Sequence:** Steps are arranged logically, usually top-to-bottom or left-to-right, showing the progression from starting material to final product via intermediates.

Example Flowchart (Propane $\rightarrow$ Propanone):



Part 2: Planning Space for Synthesis Challenge

Instructions: Use the space below (or separate paper) to plan and draft the flowchart for the synthesis problem assigned to your group (see Activity Sheet 3). Use the Chord Diagram tool for initial planning. Remember to include all intermediates, reagents, and conditions.

Assigned Problem: _____

Planning Notes (Routes explored, key steps):

Draft Flowchart:

Justification Notes (Why was this route/these steps chosen?):